

JEE MAIN REVISION MASTER: UNITS & MEASUREMENT AND DIMENSIONS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Dimensional Analysis & Physical Constants	5 PYQs	Dimensions & Units
Moderate	Errors in Measurement & Vernier / Screw Gauge	4 PYQs	Experimental Physics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Dimensional Analysis & Physical Constants	$[h] = [ML^2T^{-1}]$, $[G] = [M^{-1}L^3T^{-2}]$, $\left[\frac{1}{\sqrt{\mu_0\epsilon_0}}\right] = [LT^{-1}]$
Errors in Measurement & Vernier / Screw Gauge	$\text{LC} = 1 \text{ MSD} - 1 \text{ VSD}$, $Z = A^a B^b \implies \frac{\Delta Z}{Z} = a \frac{\Delta A}{A} + b \frac{\Delta B}{B}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Dimensional Analysis & Physical Constants (Moderate Yield — 5 PYQs)

Core Principle: Dimensional analysis of physical constants and equations. Determining dimensions of ϵ_0 , μ_0 , h , G and testing homogeneity of physical equations.

■ **Caution / JEE Pitfall:** Confusing angular momentum $[ML^2T^{-1}]$ with linear momentum $[MLT^{-1}]$, or torque with work (same dimensions $[ML^2T^{-2}]$ but different physical natures).

■ **Speed Trick:** In KCET, use the speed of light relation $c = 1/\sqrt{\mu_0\epsilon_0}$ to find dimensions of $(\mu_0\epsilon_0)^{-1/2}$ in 2 seconds without computing μ_0 or ϵ_0 separately.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Match the physical quantities given in List-I with dimensions expressed in terms of mass (M), length (L), time (T) and electric current (A) ..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Which of the following expression can be deduced on the basis of dimensional analysis? (All symbols have their usual meanings)..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Dimensional formula for activity of a radioactive substance is..." [Ans: C]

3.2 Errors in Measurement & Vernier / Screw Gauge (Moderate Yield — 4 PYQs)

Core Principle: Calculation of least count for Vernier calipers (1 MSD - 1 VSD) and Screw Gauge (Pitch / Total divisions). Error propagation in products and powers: $(\Delta Z / Z) = a(\Delta A / A) + b(\Delta B / B)$.

■ **Caution / JEE Pitfall:** When calculating percentage error, remember that powers multiply the fractional error: if $Z = A^3$, the error in A is multiplied by 3!

■ **Speed Trick:** Always check zero error: Positive zero error must be subtracted from the observed reading; negative zero error must be added.

Representative Source PYQs:

- **[2023 Shift] Q#4 (MCQ):** "The true length of a wire is 3.678cm . When the length of this wire is measured using instrument A , the length of the wire is 3.5cm" **[Ans: N/A]**
- **[2022 Shift] Q#1 (MCQ):** "The Vernier scale of a travelling microscope has 50 divisions which coincides with 49 main scale divisions. If each main scale division is \$..." **[Ans: N/A]**
- **[2020 Shift] Q#2 (MCQ):** "A cylindrical wire has a mass $(0.3 \pm 0.003)g$, radius $(0.5 \pm 0.005)mm$ and length $(6 \pm 0.06)cm$. The maximum percentage error in ..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Match the physical quantities given in List-I with dimensions expressed in terms of mass (M), length (L), time (T) and electric current (A) given in List-II. List-I List-II (a) Torque (i) $[M^{-1}L^{-2}T^4A^2]$ (b) Gravitational constant (ii) $[M^1L^2T^{-1}]$ (c) Capacitance (iii) $[M^{-1}L^3T^{-2}]$ (d) Planck's constant (iv) $[M^1L^2T^{-2}]$ Codes:

- (A) a - iv, b - ii, c - iii, d - i
- (B) a - iv, b - iii, c - i, d - ii
- (C) a - iv, b - i, c - iii, d - ii
- (D) a - ii, b - i, c - iii, d - iv

Correct Answer: None

Solution Analysis: For each quantity, we write its dimensional formula. Torque is given by Torque=Force $\times$ distance Force has dimensions $[MLT^{-2}]$ So, $[\text{Torque}] = [MLT^{-2}][L] = [ML^2T^{-2}]$ So, torque matches (iv). For gravitational constant G , from Newton's law of gravitation, $F = \frac{Gm_1m_2}{r^2}$ So, $G = \frac{Fr^2}{m_1m_2}$ Now, $[G] = \frac{[MLT^{-2}][L^2]}{[M^2]}$ $[G] = [M^{-1}L^3T^{-2}]$ So, gravitational constant matches (iii). Capacitance is $C = (Q / V)$ Charge has dimensions $[Q] = [AT]$ Potential difference is $V = \frac{\text{Work}}{\text{Charge}}$

Benchmark Problem 2: KCET 2023 (Q#4)

The true length of a wire is 3.678cm . When the length of this wire is measured using instrument A , the length of the wire is 3.5cm . When the length of the wire is measured using instrument B , it is found to have length 3.38cm . Then, the

- (A) measurement with A is more accurate and precise.
- (B) measurement with A is more accurate while measurement with B is more precise.
- (C) measurement with B is more accurate and precise.
- (D) measurement with A is more precise while measurement with B is more constant.

Correct Answer: None

Solution Analysis: Given, actual length of wire (l_0)=3.678cm Measurement with instrument, A=3.5cm Measurement with instrument, B=3.38cm Measurement of A is close to l_0 so it is more accurate but measurement of B is more precise as it can count upto two decimal places.